



MATS

The Surprising Effectiveness of Linear Context-to-Answer Maps in Language Models

A context-conditioned metamodel of answer activations

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TL;DR

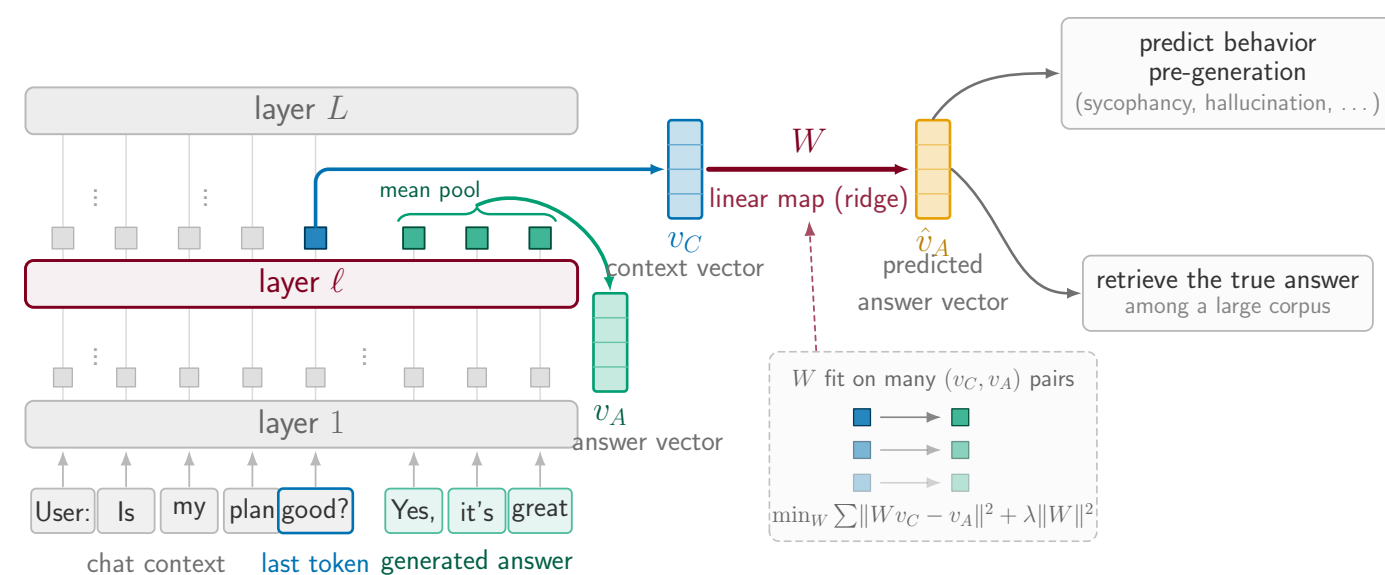
- A single **linear map** predicts an LLM’s mean answer activation from its final context activation **before generation**: held-out R^2 **0.75**, and rank-1 retrieval of the true answer among 10,000 candidates at **0.98**
- The base model already reaches **67% of the fully post-trained R^2** ; SFT closes most of the rest, and DPO and RLVR barely change it
- It is **shared between the assistant in chat template, assistant in stories, and other fictional characters in stories**, providing evidence for the persona selection model
- It can be used to predict **sycophancy, hallucination and evil before inference**.

MOTIVATION

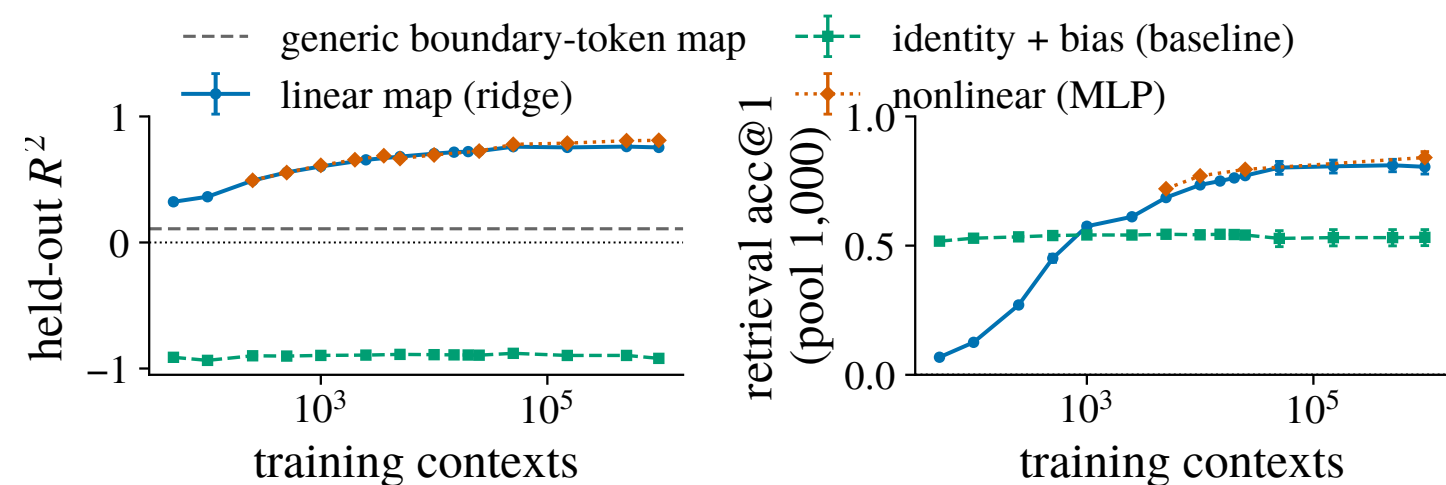
- LLMs can be viewed as deterministic maps from *contexts* to *answer distributions*
- A **simple approximation** of that map would be extremely useful: much easier to analyze theoretically, sample from, and understand the effects of finetuning
- But it need not exist: the map is billions of parameters applying dozens of nonlinear layers.
- Prior work either uses simple but *narrow* predictors (specific tokens, activations, behaviors), or distillation, a complex whole-distribution predictor
- Our question**: can a **simple** approximation capture the mapping *as a whole*?

METHODOLOGY

- Context** = the full model input (persona + question); **answer** = one sampled completion.
- Read one layer’s residual stream: **context vector** v_C at the last context token, **answer vector** v_A averaged over the answer’s tokens.
- Fit one linear map $v_A \approx Wv_C$ (ridge) over many pairs; score on **contexts it never saw**.
- Metrics: held-out R^2 ; **retrieval** (is the true answer the prediction’s nearest neighbour among a large corpus); **behavior prediction** (LLM judged)

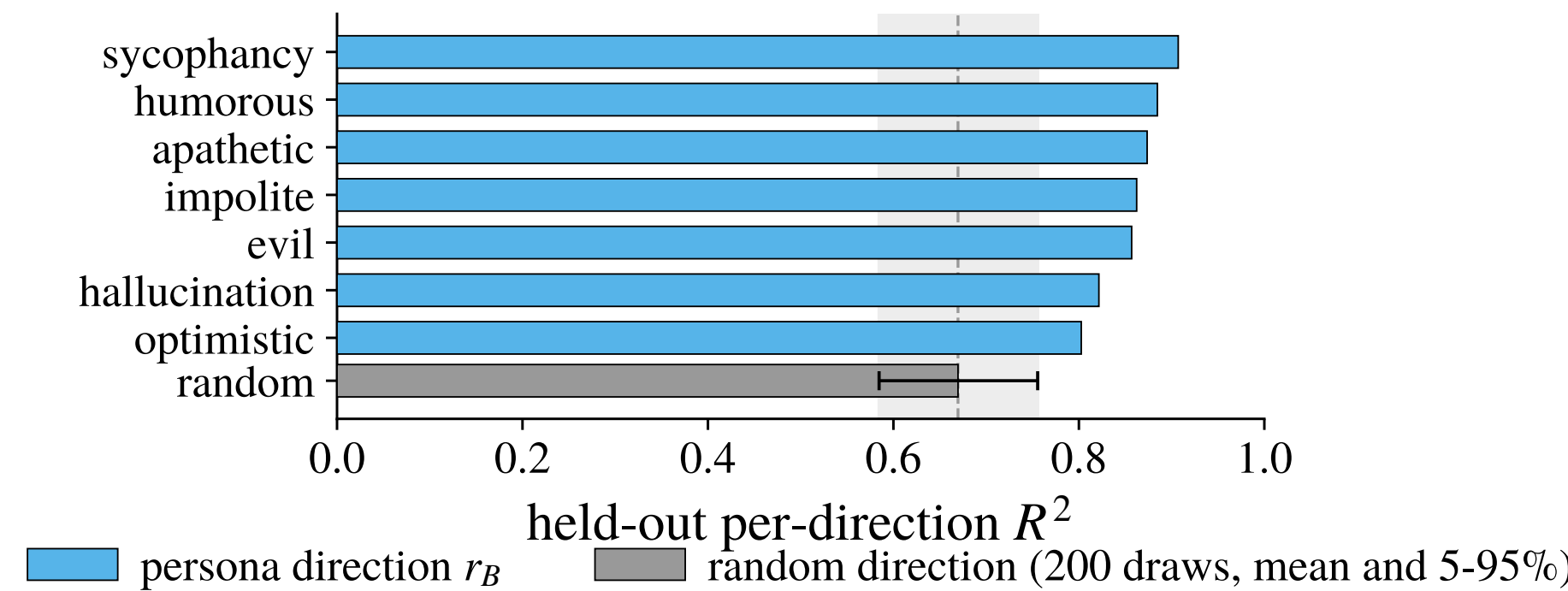


1. THE MAPPING HAS HIGH PREDICTIVE POWER AND IS MOSTLY LINEAR



- The mapping has high predictive power**: held-out R^2 0.75, rank-1 retrieval 0.98 at 1 million contexts
- The mapping is mostly linear**: the nonlinear (MLP) map adds only ~ 0.06 R^2 over the linear map, and only at large n (potentially this advantage would grow as you scale up the number of contexts)
- The mapping is not present for generic boundary tokens and is not just copying the context vector (baseline)**
- We study the **linear** map here and leave more complex maps to future work

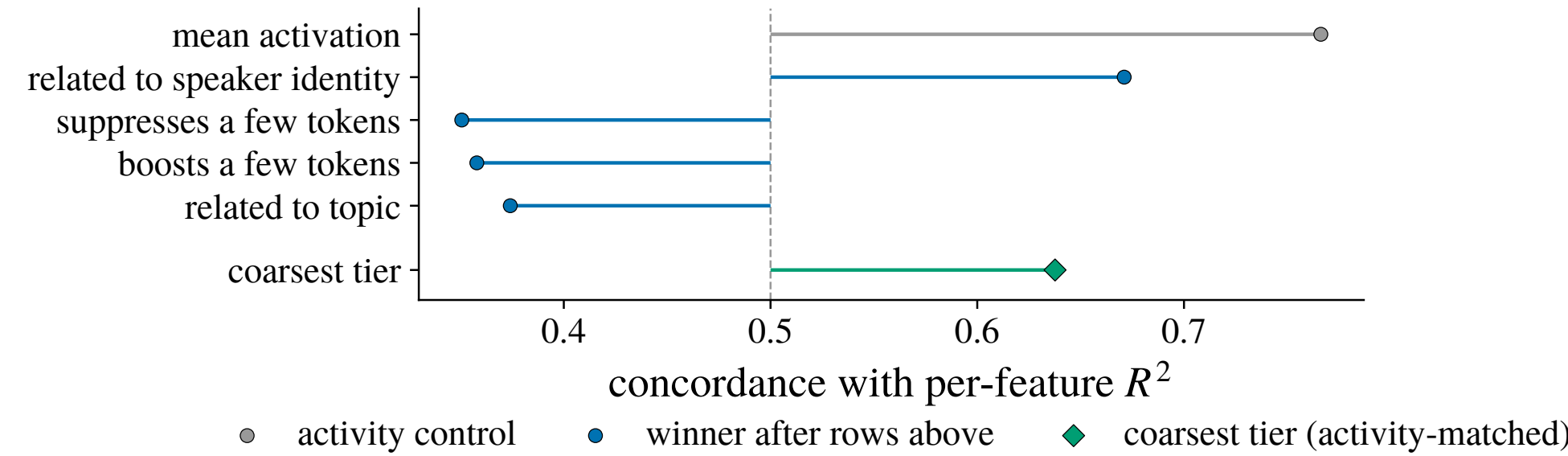
2. THE MAPPING PREDICTS USEFUL PARTS OF THE ANSWER



- Persona-vector directions are among the best-predicted directions of the answer**: R^2 **0.80–0.91** against a random-direction mean of **0.67**, and every one does better than 95% of random directions (0.76)

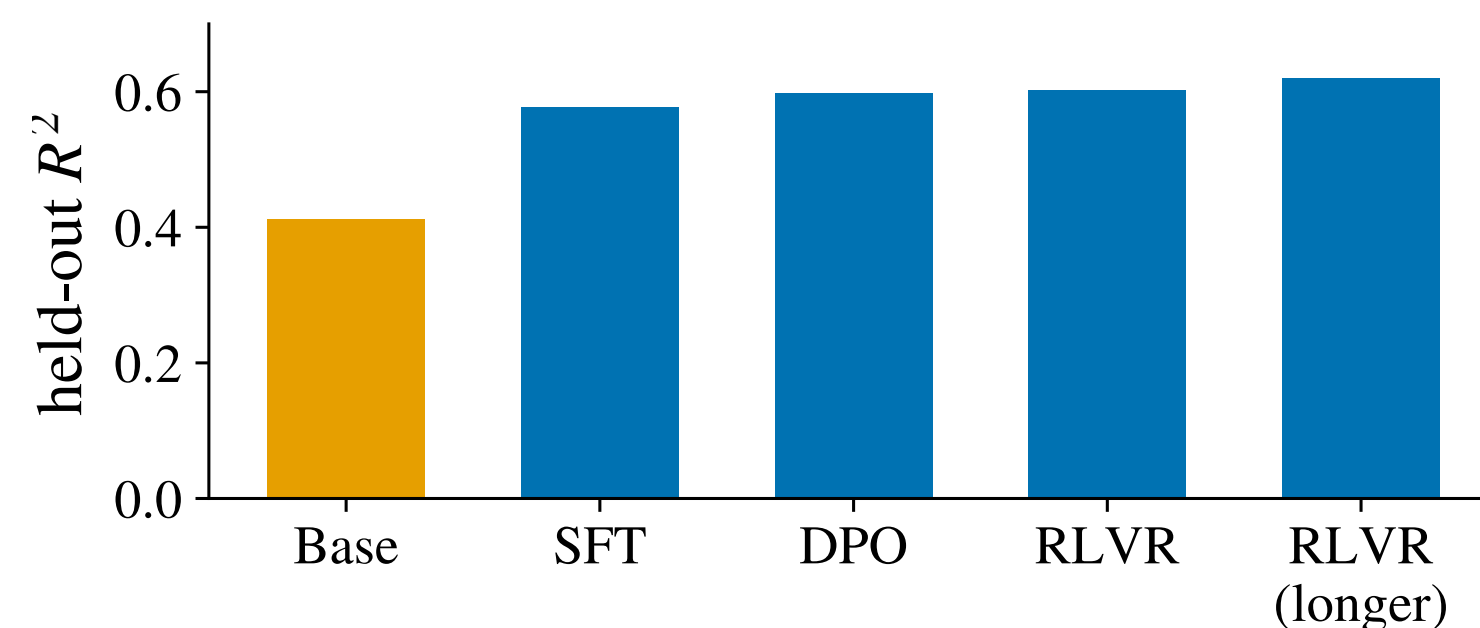
3. THE MAPPING IS BETTER AT PREDICTING HIGH-LEVEL AND IDENTITY RELATED FEATURES OF THE ANSWER

We train another map from the context to the answer’s **SAE features**, then partial out predictors *one at a time* to see which survives.



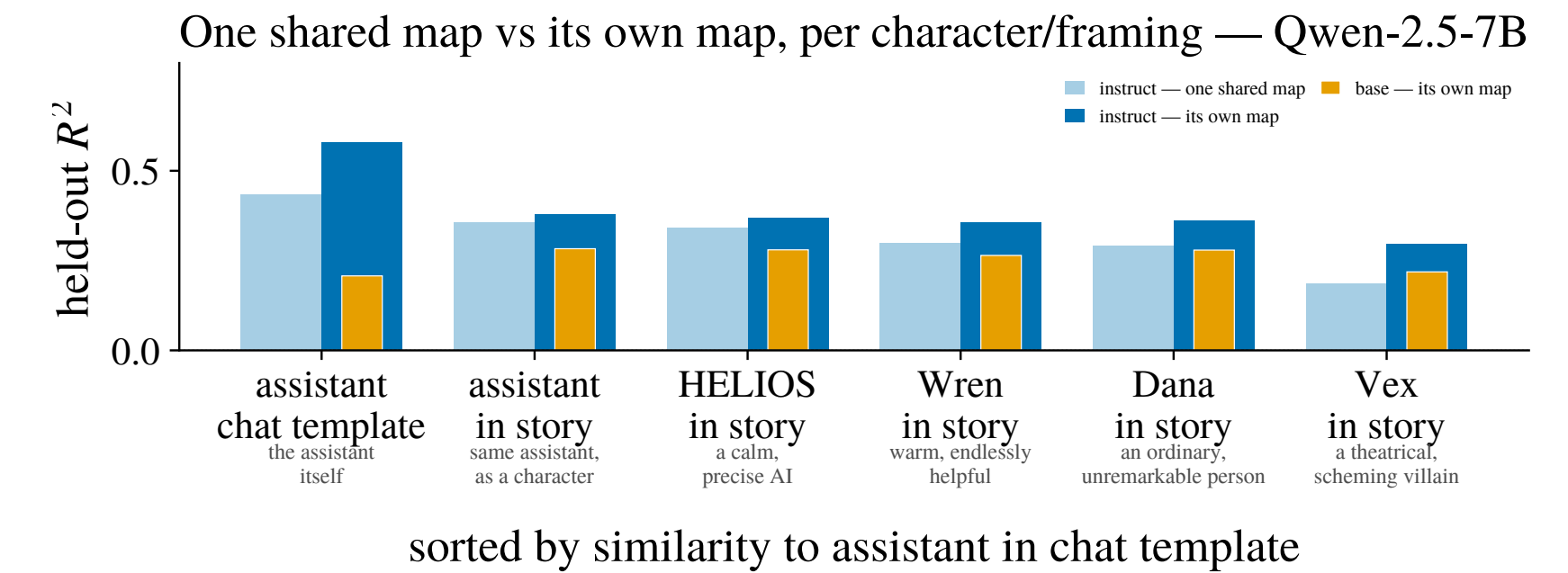
- Controlling for mean activation, the best predictors are
 - being related to speaker identity
 - not suppressing/promoting a few tokens
 - not being related to topic
 - being in the coarsest tier of a Matryoshka SAE (intuitively, being high-level)
- Therefore, the mapping seems to be
 - good at predicting **high-level** and **identity-related** features
 - bad at predicting **low-level** and **topic-related** features
 - suggests it might be some kind of **persona mapping**

5. THE MAPPING IS ALREADY LARGELY PRESENT IN THE BASE MODEL AND REFINED THROUGHOUT POST TRAINING



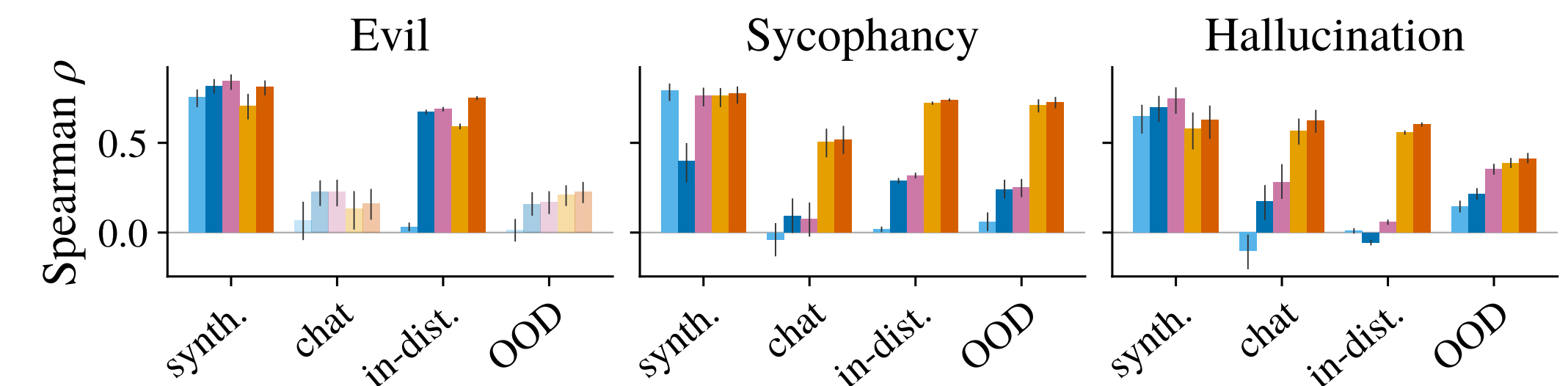
- The mapping is already largely present in the base model**: R^2 0.41 before any post-training, **67%** of the 0.62 the fully post-trained model reaches.
- SFT has the largest effect on the mapping** (+0.17), with DPO and both RLVR stages together adding only +0.04.

6. THE MAPPING IS SHARED WITH STORY CHARACTERS



- One shared map does almost as well as per-character/fragment maps**: 62-94% of each framing/character’s own-map performance
 - This suggests the model is representing the assistant in the chat template similarly to a character in a story, providing evidence for the **Persona Selection Model**
- Characters more similar to the assistant have better mappings**
 - But this difference only arises in the instruct model: the assistant’s privileged position/representation as a persona comes from **post-training**

7. APPLICATION: BEHAVIOR PREDICTION BEFORE INFERENCE



- persona vector on the context
- persona vector on the mapped answer
- regression on the context
- regression on the mapped answer
- persona vector on the real answer (oracle)
- faded = behavior almost never occurs here

- Persona vectors projected on the context vector fail to generalize to realistic contexts**
 - Projecting persona vectors on the mapped answer does almost as well as projecting on the real answer
- Regression on the mapped answer does the best of all methods**
 - This showcases one potential application of this mapping

OTHER RESULTS

- The mapping does not get reliably better with model scale**
- In thinking models, the mapping predicts just as well from the pre-CoT context vector and the post-CoT context vector**
 - suggesting that whatever is predicted by the mapping is not affected by CoT

IMPLICATIONS

- A linear mapping can approximate much of the context-answer mapping implemented by a LLM
- Applications: predicting behavior, automated redteaming, predicting effect of fine-tuning
- Evidence for persona selection model
- Motivates further work into metamodels of answer activations and the linearity in transformers

FUTURE WORK (MOSTLY IN PROGRESS)

- Trying more complex estimators (diffusion, flow matching, etc.)
- Predicting the entire answer distribution to bound probability of rare behaviors
- Theoretical analysis of the mapping
- Predicting other properties of the answer (correctness, refusal, etc.)
- Predicting the effect of fine-tuning on the mapping/model
- Automated redteaming / jailbreaking.
- Single-token next-activation mapping

REFERENCES [1] Chen et al., Persona vectors, 2025; [2] Betley et al., Emergent misalignment, 2025; [3] Marks et al., The persona selection model, 2026; [4] Lu et al., The assistant axis, 2026; [5] Hosseini & Fedorenko, Trajectory straightening, 2023; [6] Park et al., The linear representation hypothesis, 2024.